

Business Recommendation on Advertising Investment Plan

Business Recommendation

Objective

The purpose of this analysis was to identify the most effective allocation of digital advertising expenditure using Google's Meridian Marketing Mix Modeling(MMM) framework. The model evaluated four Google advertising channels—Google Paid Search, Google Shopping, Google PMAx, and Google Display—to understand their contribution to purchases and determine how future marketing budgets should be allocated.

Key Findings

The results show that the current marketing budget can be used more efficiently by concentrating investment on the highest-performing channels rather than distributing spend evenly across all campaigns.

Among the four channels, Google Paid Search and Google Shopping consistently produced the strongest results. Both channels generated the largest contribution to purchases and continued to show positive returns as investment increased. In contrast, Google PMAx displayed signs of diminishing returns, indicating that additional spending produced smaller incremental gains. Google Display contributed only a small proportion of the overall performance and therefore offered limited value compared with the other channels.

The budget optimization also demonstrated that reallocating spend can improve marketing performance without increasing the total budget. Redirecting investment from lower-performance channels toward Paid Search and Shopping resulted in higher predicted purchases and improved overall efficiency.

Scenario Analysis

Four investment scenarios were evaluated to assess different budget strategies.

The 20% budget increase generated the highest predicted number of purchases(approximately 49,176), showing that additional investment can be effective when allocated using Meridian's optimization.

Increasing investment in Google Paid Search by 50% produced the strongest allocation for a fixed budget, confirming that Paid Search remains the most valuable channel for future investment.

Reducing expenditure on saturated channels, particularly Google PMAx and Google Display, improved overall return by reallocating resources toward higher-performing campaigns.

The random allocation scenario produced lower overall performance than the optimized strategies, demonstrating that marketing budgets should be allocated based on model insights rather than arbitrary adjustments.

Recommended Investment Strategy

Based on the model results, the recommended advertising allocation is:

Channel	Recommended Allocation
Google Shopping	~40%
Google Paid Search	~40%
Google PMAx	~20%
Google Display	Less than 1%

The allocation reflects the balance between maximizing predicted purchases and avoiding unnecessary spending on channels that have already reached diminishing returns.

If additional marketing budget becomes available, the preferred strategy is to increase the overall budget and let the optimization model determine the allocation, rather than distributing the additional spending equally across all channels.

Conclusion

The Meridian marketing Mix model provided clear evidence that advertising effectiveness varies considerably across channels. Concentrating investment on Google Paid Search and Google Shopping while reducing spending on saturated channels is expected to improve campaign performance and generate a greater number of purchases from the available budget.

Although the model was developed using purchase volume rather than revenue, the recommendations provide a practical framework for improving future advertising decisions. As more business data becomes available, particularly revenue per purchase, the same modelling approach can be extended to optimise marketing investment based on financial return as well as customer acquisition.

Limitations

The analysis has several limitations. The model was built using purchases as the primary KPI rather than revenue, so the evaluation focuses on predicted purchases and ROI instead of direct revenue uplift. In addition, the final model was trained using the territory with the most complete historical data because the available territories did not share a consistent time series. Finally, the model assumes that historical relationships between advertising spend and customer behaviour remain stable over the analysis period.